

✓ Practical No 06

1. Python Implementation of the above algorithm using the scikit learn library

```
from sklearn.cluster import AgglomerativeClustering
import numpy as np
X = np.array([[1, 2], [1, 4], [1, 0],
              [4, 2], [4, 4], [4, 0]])
clustering = AgglomerativeClustering(n_clusters=2).fit(X)
print(clustering.labels_)
```

↔ [1 1 1 0 0 0]

Implementation code for the Distance Matrix Comparison

```
import numpy as np
from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt
X = np.array([[1, 2], [1, 4], [1, 0],
              [4, 2], [4, 4], [4, 0]])
```

```
Z = linkage(X, 'ward') # Ward Distance
```

```
dendrogram(Z) #plotting the dendrogram
```

```
plt.title('Hierarchical Clustering Dendrogram')
plt.xlabel('Data point')
plt.ylabel('Distance')
plt.show()
```

↔ Hierarchical Clustering Dendrogram



